

RAVI RANJAN PRASAD

Full Stack Chatbot Developer | Generative AI & backend Engineer

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PROFESSIONAL SUMMARY

Full Stack Chatbot Developer and AI Engineer with **6+ years of experience** designing, developing, and supporting **enterprise conversational AI applications and high-concurrency backend services**. Experienced in building **Generative AI assistants end to end**, including **LLM integration, prompt engineering, Retrieval-Augmented Generation (RAG), knowledge retrieval, and conversation-state management**. Strong backend expertise in **Python (FastAPI), Node.js, REST APIs, MongoDB, Redis, and microservices**, with hands-on experience deploying applications on **Microsoft Azure (App Service, Azure Functions)** and implementing **GitHub Actions CI/CD pipelines**. Experienced in integrating enterprise platforms including **ServiceNow (Incident Management, Knowledge Base, Service Catalog), Genesys Live Chat, and Multi-tenant architecture with Microsoft Teams, Zoom, Google Chat, etc.** along with **application monitoring, performance optimization, production troubleshooting, and zero-downtime deployments**.

TECHNICAL SKILLS

Languages: JavaScript, TypeScript, Python, SQL

backend & Frameworks: Node.js, NestJS, Express, FastAPI, REST API design, Microsoft Bot Framework, microservices, asynchronous processing

Generative AI / Conversational AI: LLM integration (Azure OpenAI, GPT-4.1 & mini), prompt engineering, RAG pipelines, vector search and embeddings, agentic / multi-turn dialogue design, intent handling, fallback and escalation logic, chatbot evaluation metrics

Cloud & DevOps: Microsoft Azure (App Service, Azure Functions, Azure OpenAI, Azure AI Search, Cosmos DB, Application Insights, Key Vault), CI/CD pipeline design, GitHub Actions, Azure DevOps Pipelines, Docker, Git, blue-green and zero-downtime deployments

Databases & Caching: MongoDB, Cosmos DB, Redis, query optimization, indexing, caching strategy

Enterprise Integrations: ServiceNow (Incident/Ticketing, Knowledge Base, Interactions, Service Catalog), Genesys Live Chat, Microsoft Teams, Zoom API, Azure AD authentication, webhooks

Practices: Agile/Scrum, code reviews, unit and integration testing, observability and logging, production support, technical documentation, knowledge transfer

PROFESSIONAL EXPERIENCE

Full Stack Chatbot Developer / AI Backend Engineer | Tata Consultancy Services (TCS) Oct 2021 – Present

Client: Exxon Mobil Corporation — Enterprise AI Assistant (ITChat) Sep 2024 – Present

- Developed and maintained **backend services for an enterprise Generative AI assistant** using **Python (FastAPI) and Node.js**, supporting high-concurrency conversational workloads for global IT support.
- Engineered **GenAI workflows** covering **LLM integration, prompt engineering, response grounding, conversation context/state management, and fallback handling** for low-confidence responses.
- Designed and enhanced **Retrieval-Augmented Generation (RAG) pipelines** over enterprise knowledge sources such as **policy documents, IT runbooks, and knowledge base articles**, including document chunking, embedding refresh, retrieval optimization, and relevance tuning.
- Integrated **ServiceNow REST APIs** for **incident creation, ticket status lookup, and service catalog requests**, enabling IT transactions directly through **Microsoft Teams**.
- Deployed backend services on **Microsoft Azure (App Service, Azure Functions)** and implemented **GitHub Actions CI/CD** for automated build, test, and deployment workflows.
- Implemented **Azure Application Insights** telemetry and optimized application performance through **caching, query tuning, and asynchronous processing**, monitoring latency, error rates, and fallback behavior.
- Collaborated with **QA teams and client stakeholders** across **UAT, production releases, defect resolution, and post-release validation**.

Client: Shell plc — IT Support Chatbot (Sam)

Oct 2021 – Sep 2024

- Developed and enhanced an enterprise **IT support chatbot using Microsoft Bot Framework and Node.js**, supporting **multi-turn conversations, context retention, Adaptive Cards, and fallback handling** across Web and Microsoft Teams.
- Implemented **Natural Language Processing (NLP)** capabilities using **Azure Conversational Language Understanding (CLU)** to identify user intent and route customer utterances to appropriate chatbot flows.
- **Trained and maintained Azure CLU models using customer utterances**, supporting intent recognition and conversational routing for IT support scenarios.
- Integrated **ServiceNow Incident/Ticketing, Knowledge Base, and Interactions** with **Genesys Live Chat**, enabling automated support workflows and **bot-to-agent escalation**.
- Developed and maintained **REST APIs and backend services**, including **MongoDB query optimization and Redis caching** for session and frequently accessed data.
- Implemented **structured logging, environment-specific configuration, and secret management** to improve application traceability and operational support.
- Collaborated with **client product owners** to translate support requirements into conversational flows and used **containment and escalation metrics** to identify chatbot improvement areas.
- Supported **production troubleshooting and root-cause analysis**, resolving live defects and implementing permanent fixes.

Associate Software Engineer | Wipro

Jun 2019 – Jun 2020

Enterprise Asset Management- AIRFORCE INDIA (IAF-Maximo) & Core Integration Stream (IBF-AIC)

- Supported **IBM Maximo Enterprise Asset Management (EAM)** platform using **Java-based application components and SQL/MySQL databases** for large-scale asset tracking and maintenance operations.
- Optimized **SQL queries and database operations**, troubleshooting performance issues, resolving data integrity problems, and improving reliability of backend data processing.
- Contributed to **Maximo application development and maintenance**, including defect resolution, application enhancements, and backend troubleshooting across enterprise workflows.
- Supported **system integration validation** across Maximo and connected enterprise applications, analyzing data flows, identifying integration defects, and validating fixes.
- Participated in **Agile development sprints**, collaborating with development and QA teams on requirements, implementation, testing, defect resolution, and production-ready releases.

EDUCATION

B.Tech, Computer Science & Engineering | Chandigarh University, Mohali

2015 – 2019

CERTIFICATIONS & ACHIEVEMENTS

- Microsoft Certified: Azure Fundamentals (**AZ-900**)
- Microsoft Certified: Azure AI Fundamentals (**AI-901**)
- Claude Certified **Architect** – Foundations
- Claude Certified **Developer** – Foundations
- **Tech 4.0** Certified Professional — TCS internal competency framework
- Certification training in Linux Administration and Networking Fundamentals
- Recipient of multiple **TCS Digital Awards** and **client appreciation** certificates for project delivery